

32
33

Knowledge:

1. Be sure to show your work for the following questions...Do NOT just state the final answer.

a) Simplify $(e^{\ln(3x)})(\ln(e^{2x}))$

b) Solve for x: $\ln(e^{3-x}) = 8$

[2]

$$\begin{aligned} &= 3x(2x \ln e) \\ &= 3x(2x) \\ &= 6x^2 \end{aligned}$$

$$(3-x) \ln e = 8$$

$$3-x = 8$$

$$x = -5$$

$$x = -5$$

2. Determine the derivative for the following functions. Express final answers in a fully factored form.

a) $y = 12^{-4x+9} \cos(x^2)$

b) $y = \frac{\sin x}{1 + \cos x}$

$$\frac{dy}{dx} = 12^{-4x+9} \ln(12) \cos(x^2) + 12^{-4x+9} (-\sin(x^2)) (2x)$$

$$\frac{dy}{dx} = \frac{\cos x (1 + \cos x) - \sin x (-\sin x)}{(1 + \cos x)^2}$$

[5]

$$= 2 \cdot 12^{-4x+9} (-2 \ln(12) \cos(x^2) - x \sin(x^2))$$

$$= (-2)(12^{-4x+9}) (2 \ln(12) \cos(x^2) + x \sin(x^2))$$

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$$

$$= \frac{\cos x + 1}{(1 + \cos x)^2}$$

$$= \frac{1}{1 + \cos x}$$

c) $y = \frac{\log_{10} x}{10^x}$

$$\frac{dy}{dx} = \frac{\frac{1}{x \ln 10} (10^x) - \log x (10^x) \ln 10}{(10^x)^2}$$

$$\frac{\log x \cdot 10^{-x}}{10^x} = \frac{10^{-x}}{x \ln 10} + \log x \cdot 10^{-2x} \ln 10$$

$$= \frac{10^{-x}}{x \ln 10} - \frac{\log x \ln 10}{10^x}$$

$$= \frac{10^{-x} (10^x) - \log x \ln 10 (x \ln 10)}{x^2 10^{2x} \ln 10}$$

$$= \frac{1 - x \log x (\ln 10)^2}{x^2 10^{2x} \ln 10}$$

$$= \frac{1 - x \log x (\ln 10)^2}{x^2 10^{2x} \ln 10}$$

[3]

2

$$= \frac{10^x}{x \ln 10} - \log x \cdot 10^x \ln 10$$

$$= \frac{10^x x^{-1} (\ln 10)^{-1} - \log x \cdot 10^x \ln 10}{10^{2x}}$$

(-1) exponent

$$= \frac{10^x \ln 10^{-1} (x^{-1}) (1 - \log x (\ln 10)^2)}{10^{2x}}$$

$$= \frac{1 - x \log x (\ln 10)^2}{10^{2x} \ln 10}$$

9

Communication

1. Michael was asked to derive and fully factor his final answer. His solution is incorrect, but he cannot find his mistake(s). Explain where he made his mistake(s) and correct them.

[2]

When he derived e^{-2x} , he forgot to derive the exponent, it should be like this:

$$y = 3x^2 e^{-2x}$$

$$\frac{dy}{dx} = 6x e^{-2x} + 3x^2 e^{-2x} (-2)$$

$$\frac{dy}{dx} = 6x e^{-2x} - 6x^2 e^{-2x}$$

$$\frac{dy}{dx} = 6x e^{-2x} (1 - x)$$

Also he didn't factor out x .

2. In two ways, explain why the derivative of $\sin^2 x + \cos^2 x$ is zero.

[3]

Firstly, $\sin^2 x + \cos^2 x$ is equivalent to 1, and the derivative of a constant is zero:

$$f(x) = \sin^2 x + \cos^2 x$$

$$f'(x) = 2 \sin x \cos x + 2 \cos x (-\sin x)$$

$$f'(x) = 2 \sin x \cos x - 2 \sin x \cos x$$

$$f'(x) = 0$$

Secondly, if you do the math, and derive:

Application:

1. Determine the exact points where the function $g(x) = x^4 e^{-x}$ could have points of inflection.

[4]

$$g'(x) = 4x^3 e^{-x} + x^4 e^{-x} (-1)$$

$$= 4x^3 e^{-x} - x^4 e^{-x}$$

$$g''(x) = 12x^2 e^{-x} - 4x^3 e^{-x} - 4x^3 e^{-x} + x^4 e^{-x} (-1)$$

$$g''(x) = 12x^2 e^{-x} - 8x^3 e^{-x} + x^4 e^{-x} (-1)$$

$$g''(x) = x^2 e^{-x} (12 - 8x + x^2)$$

$$g''(x) = x^2 e^{-x} (x^2 - 8x + 12)$$

$$x = 0, 6, 2$$

$$g(0) = 0$$

$$g(2) = 16 e^{-2} = \frac{16}{e^2}$$

$$g(6) = 1296 e^{-6} = \frac{1296}{e^6}$$

$$A(0, 0)$$

$$B(2, \frac{16}{e^2})$$

$$C(6, \frac{1296}{e^6})$$

2. Find the parameter a such that the function $y = \sin(ax)$ satisfies the relation: $y + 3y'' = 0$

[3]

$$y = \sin(ax)$$

$$y' = a \cos(ax)$$

$$y'' = -a^2 \sin(ax)$$

$$\sin(ax) + 3(-a^2 \sin(ax)) = 0$$

$$\sin(ax) - 3a^2 \sin(ax) = 0$$

$$\sin(ax)(1 - 3a^2) = 0$$

$$\sin(ax) = 0$$

$$ax = 0, \pi, 2\pi$$

$$a = 0, \frac{\pi}{x}, \frac{2\pi}{x}$$

$$1 - 3a^2 = 0$$

$$3a^2 = 1$$

$$a^2 = \frac{1}{3}$$

$$a = \pm \sqrt{\frac{1}{3}}$$

$$a = \frac{n\pi}{x} \text{ or } \sqrt{\frac{1}{3}}, -\sqrt{\frac{1}{3}}$$

where n is any whole number integer.

3. Terrific Toys Company finds its monthly profit P in dollars is approximated by the function

$$P(x) = (x-10)^2 e^{x-10}$$

where x is the amount in hundreds of dollars spent on advertising each month. When is the company's

monthly profit decreasing? when $P'(x) < 0$, $M < 0$, and $P(x) \downarrow$

$$P(x) = (x-10)^2 e^{x-10}$$

$$P'(x) = 2(x-10)e^{x-10} + (x-10)^2 e^{x-10}$$

$$= (x-10)e^{x-10}(2 + x-10)$$

$$= (x-10)e^{x-10}(x-8)$$

$$C.P.: 8, 10$$

$$(-\infty, 8) \quad (8, 10) \quad (10, \infty)$$

$$S'(x) \quad + \quad - \quad +$$

$$f(x) \quad \uparrow \quad \downarrow \quad \uparrow$$

Therefore the company's monthly profit is decreasing when between \$800 and \$1000 is spent on advertising each month.

4. Determine when the function, $y = -2\cos(x - \frac{\pi}{6})$, has a horizontal tangent on the interval of $-2\pi \leq x \leq 0$. Provide exact solutions.

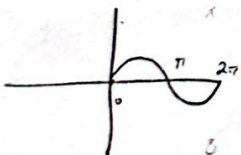
$$y = -2\cos(x - \frac{\pi}{6})$$

$$[-2\pi, 0]$$

$$y' = -2(-\sin(x - \frac{\pi}{6}))(1)$$

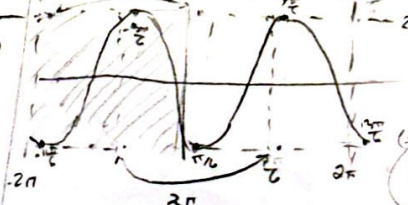
$$= 2\sin(x - \frac{\pi}{6})$$

$$0 = \sin(x - \frac{\pi}{6})$$



$$(x - \frac{\pi}{6}) = 0, \pi, 2\pi$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}$$



$$x = \frac{\pi}{6} - 2\pi = \frac{\pi}{6} - \frac{12\pi}{6} = -\frac{11\pi}{6}$$

$$x = \frac{7\pi}{6} - 2\pi = \frac{7\pi}{6} - \frac{12\pi}{6} = -\frac{5\pi}{6}$$

This function has horizontal tangents at $x = -\frac{11\pi}{6}$ and $x = -\frac{5\pi}{6}$ on the interval $-2\pi \leq x \leq 0$.

Thinking: [4 Marks]

1. Determine the x -value(s) of all local extrema on the function where

$$f(x) = 2\cos x - 2\sin 2x, \text{ on}$$

$$0 \leq x \leq 2\pi$$

$$f'(x) = -2\sin x - 4\cos 2x$$

$$= -2(\sin x + 2\cos 2x)$$

$$= -2(\sin x + 2(1 - 2\sin^2 x))$$

$$= -2(\sin x + 2 - 4\sin^2 x)$$

$$= -2(-4\sin^2 x + \sin x + 2)$$

$$a = -4$$

$$b = 1$$

$$c = 2$$

$$\sin x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

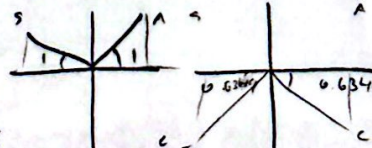
$$= \frac{-1 \pm \sqrt{1 - 4(-4)(2)}}{2(-4)}$$

$$\sin x = \frac{-1 \pm \sqrt{33}}{-8}$$

$$\sin x = 0.593, 0.843$$

$$\sin^{-1}(0.843) = x = 1.1$$

$$\sin^{-1}(0.593) = x = 0.6349$$



$$x = 1, 2.14, 0, 2\pi$$

$$3.77, 5.65$$

$$f(0) = 2$$

$$f(1) = -0.74$$

$$f(2.14) = 0.74$$

$$f(3.77) = -3.52$$

$$f(5.65) = 3.52$$

$$f(2\pi) = 2$$

12